
Sensitivity analysis for the mixed-effects model

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This script simulates seizures following missed ASM doses for a specified probability p , to determine the sensitivity of a mixed-effects approach to looking at the relationship between seizure occurrence and missed ASMs.

Format data for excel spreadsheet

Load file

```
filename = 'all_patient_diaries.xlsx';

% Extract information about number of sheets in the excel
[~, sheetNames] = xlsfinfo(filename);

% Load the data from each sheet/participant as a separate variable
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    varName = matlab.lang.makeValidName(sheetNames{i});
    assignin('base', varName, sheetData);
end
```

Extract the ASM data for each subject

End of data format will be 3D matrix [Dose X ASM X Subject]

```
% Patient 4165: Column 3 Keppra, Column 4 Epidiolex, Column 5 Fluramine
ASM(:,1:3,1)=BIDMC4165(:,3:5);

% Patient 4216: Column 3 CBD oil, Column 4 Lamotrigine
ASM(:,1:2,2)=BIDMC4216(:,3:4);

% Patient 4251: Column 3 CBD oil, Column 5 Onfi
ASM(:,1,3)=BIDMC4251(:,3);
ASM(:,2,3)=BIDMC4251(:,5);

% Patient 4417: Column 5 Keppra, Column 6 Lamictal XR, Column 7 Vimpat
ASM(:,1:3,4)=BIDMC4417(:,5:7);
```

Sensitivity analysis for the mixed-effects model

```
% Patient 4614: Column 3 Aptiom, Column 4 Xcopri
ASM(:,1:2,5)=BIDMC4614(:,3:4);

% Patient 4684: Column 5 Phenobarb, Column 6 Lamotrigine, Column 8
% Levetiracetam
ASM(:,1:2,6)=BIDMC4684(:,5:6);
ASM(:,3,6)=BIDMC4684(:,8);

% Patient 4711: Column 3 Felbatol, Column 4 Zonisamide, Column 5 Lamictal
ASM(:,1:3,7)=BIDMC4711(:,3:5);

% Patient 4722: Column 3 Felbatol, Column 4 Onfi, Column 6 Depakote, Column
% 7 Gabapentin, Column 8 Zonisamide, Column 9 Fenfluramine
ASM(:,1:2,8)=BIDMC4722(:,3:4);
ASM(:,3:6,8)=BIDMC4722(:,6:9);

% Patient 4768: Column 3 Topomax, Column 4 Epidiolex, Column 5 Lamictal,
% Column 6 Zarontin
ASM(:,1:4,9)=BIDMC4768(:,3:6);

% Patient 4835: Column 3 Oxtellar, Column 4 Epidiolex, Column 5 Clobazam,
Column 9 Fintepla
ASM(:,1:3,10)=BIDMC4835(:,3:5);
ASM(:,4,10)=BIDMC4835(:,9);

% Patient 4982: Column 3 Lamotrigine, Column 4 Banzel, Column 5 Epidiolex
ASM(:,1:3,11)=BIDMC4982(:,3:5);

% Patient 5057: Column 4 Keppra, Column 6 Onfi
ASM(:,1,12)=BIDMC5057(:,4);
ASM(:,2,12)=BIDMC5057(:,6);

% Patient 5117: Column 4 Zonisamide
ASM(:,1,13)=BIDMC5117(:,4);

% Patient 5146: Column 3 Zonisamide
ASM(:,1,14)=BIDMC5146(:,3);

% Patient 5184: Column 3 Valproic Acid, Column 4 Keppra
ASM(:,1:2,15)=BIDMC5184(:,3:4);

% Patient 5274: Column 3 Lamictal XR, Column 6 Gabapentin, Column 7
% Fycompa
ASM(:,1,16)=BIDMC5274(:,3);
ASM(:,2:3,16)=BIDMC5274(:,6:7);

% Patient 5392: Column 3 Vimpat, Column 4 Keppra, Column 5 Depakote, Column
% 6 Zonegran
ASM(:,1:4,17)=BIDMC5392(:,3:6);

% Patient 5412: Column 3 Zonegran, Column 4 Trileptal, Column 5 Fycompa
ASM(:,1:3,18)=BIDMC5412(:,3:5);
```

```
% Patient 5507: Column 3 Briviact, Column 4 Lamotrigine, Column 5
% Epidiolex, Column 6 Xcopri
ASM(:,1:4,19)=BIDMC5507(:,3:6);

% Patient 5606: Column 3 Vimpat, Column 4 Epidiolex, Column 6 Keppra,
% Column 7 Keppra XR, Column 8 XCopri
ASM(:,1:2,20)=BIDMC5606(:,3:4);
ASM(:,3:5,20)=BIDMC5606(:,6:8);

% Patient 5698: Column 3 Keppra, Column 7 Depakote ER, Column 18 XCopri
ASM(:,1,21)=BIDMC5698(:,3);
ASM(:,2,21)=BIDMC5698(:,7);
ASM(:,3,21)=BIDMC5698(:,18);

% Patient 5708: Column 9 Keppra, Column 14 XCopri, Column 20 Lamotrigine,
% Column 21 Clobazam, Column 26 Epidiolex
ASM(:,1,22)=BIDMC5708(:,9);
ASM(:,2,22)=BIDMC5708(:,14);
ASM(:,3:4,22)=BIDMC5708(:,20:21);
ASM(:,5,22)=BIDMC5708(:,26);

% Patient 5731: Column 3 Carbamazepine, Column 4 Levetiracetam, Column 5
% Vimpat, Column 6 repeat Vimpat?? will ignore for now given no negative
% values/missed doses anyways
ASM(:,1:3,23)=BIDMC5731(:,3:5);

% Patient 5782: Column 3 Vimpat, Column 6 XCopri
ASM(:,1,24)=BIDMC5782(:,3);
ASM(:,2,24)=BIDMC5782(:,6);

% Patient 5790: Column 3 Keppra
ASM(:,1,25)=BIDMC5790(:,3);

% Patient 5799: Column 3 Keppra, Column 5 Tegretol XR
ASM(:,1,26)=BIDMC5799(:,3);
ASM(:,2,26)=BIDMC5799(:,5);

% Patient 59589: Column 3 Topamax, Column 4 Briviact, Column 5 Vimpat
ASM(:,1:3,27)=BIDMC59589(:,3:5);
```

Sensitivity Analysis

Iterate 500 times, given the probabilistic nature of simulating seizures

```
significant_missed_doses = 0; % Initialize counter for significant p-values
for missed doses
significant_ma = 0; % Initialize counter for significant p-values for MA

for iteration = 1:500

P = 0.75; %Set the probability (range 0-1)
nrows=size(ASM,1);
Sim=zeros(nrows,27);
```

```
%Simulate seizure timeseries for each subject with probability p the day
%after missed ASM doses. 1 = seizure, 0 = no seizure.
for i = 1:27
for j = 1:nrows
    %Check for times when all doses are missed
    row = ASM(j, :, i);
    non_zero=row(row~=0);
    if all(non_zero < 0)
        for k = j+1:min(j+2, nrows)
            if rand() < P
                Sim(k,i)=1;
            end
        end
    end
end
end

% As the raw data is in half days, let's shrink the temporal data such that
% it goes by days rather than half days. The below commands just takes into
% account whether there was a seizure/no seizure for a given day.
[numRows, numCols] = size(Sim);
seizure_days = zeros(numRows / 2, numCols);
for row = 1:numRows/2
    for col = 1:numCols
        % Check if there is a positive number in the two consecutive rows
        if Sim(2*row-1, col) > 0 || Sim(2*row, col) > 0
            seizure_days(row, col) = 1;
        end
    end
end

% Since we are looking at the relationship between MA and seizures on day
% sliding window + 1, the first day would be day 91 since the MA requires
% 90 days of data. Let's create a matrix of seizure days that only starts
% at day 91 for further analysis later on.
seizure_days_forecast=seizure_days(91:281, :);
```

Calculate MA for a 90 day window

Essentially within a 90 day sliding window, this takes the number of seizure days divided by 90 for an estimate of seizure risk. Note that this is done on the simulated seizure timeseries.

```
for i = 1:191
    for j = 1:27
        MA(i,j)=sum(seizure_days(i:i+89,j));
        MA_90(i,j)=MA(i,j)/90;
    end
end
```

Matrix of missed doses

The raw data for the ASM data is in half days. Let's similarly shrink the temporal resolution such that we look at ASMs in days rather than half days. We will account for missed doses in terms of "administration periods". I.e., if an

individual missed morning doses of three different ASMs, that only counts as one missed administration period. For the possible range of values here, for each day it can be 0 (no missed doses), 1 (1 missed administration period, i.e., morning vs night), and 2 (missed both morning and night doses if scheduled for it).

```
[numRows, numCols, numSubs] = size(ASM);
Missed_Doses = zeros(numRows / 2, numCols, numSubs);
    for row = 1:numRows/2
        for col = 1:numCols
            for subs = 1:numSubs
                if ASM(2*row-1, col, subs) < 0 && ASM(2*row, col, subs) < 0
                    Missed_Doses(row, col, subs) = 2;
                elseif ASM(2*row-1, col, subs) < 0 && ASM(2*row, col, subs) >= 0
                    || ASM(2*row-1, col, subs) >= 0 && ASM(2*row, col, subs) < 0
                        Missed_Doses(row, col, subs) = 1;
                end
            end
        end
    end

for j = 1:27
    Storage=Missed_Doses(:, :, j);
    Missed_Doses_Periods(:, j)=max(Storage, [], 2);
end

% Since the first 90 days are used to estimate MA, for the missed doses,
% the first number used should be on day 90 to look at how it influences
% seizure risk on day 91. The last day used should be day 280, since day
% 281 is the last day to look at seizure risk.
Missed_Doses_Periods_clean=Missed_Doses_Periods(90:280, :);
```

Organize the data for mixed effects analysis

Let's organize the data first into a table and concatenate the variables vertically. 191 estimates x 27 subjects = 5157.

```
MAConcat=reshape(MA_90, [5157, 1]); %concatenate the MA data into a vector
MissedDosesConcat=reshape(Missed_Doses_Periods_clean, [5157, 1]); %concatenate
the missed ASMs into a vector
SeizureDaysConcat=reshape(seizure_days_forecast, [5157, 1]); %concatenate the
seizure day (MA window + 1) data

%Let's create a subject ID that corresponds to each of the entries above
for i = 1:27;
    SubjectID(1:191, i)=i;
end
SubjectIDConcat=reshape(SubjectID, [5157, 1]);

% Z-score standardization of the predictor variables
MissedDosesZ = zscore(MissedDosesConcat);
MAZ = zscore(MAConcat);

% Place the data into a single table
% Table will essentially have one column consisting of Z-scored MA, Z-scored
missed doses the
% day prior, seizures on day window+1, and subject ID
```

```
tbl = table(MissedDosesZ, MAZ, SubjectIDConcat, SeizureDaysConcat,  
'VariableNames', {'MissedDoses_tbl', 'MA_tbl', 'ID_tbl', 'Seizure_tbl'});
```

Fit the mixed effects binomial model:

Seizure on day window+1 is the response variable (0 = no seizure, 1 = seizure) Z-scored missed doses the day prior is a fixed effect/predictor variable Z-scored MA for the window is a fixed effect/predictor variable Subject ID is the grouping variable/random effect Of note, we will allow for some flexibility and let the slopes between MA and missed doses vary across individuals Essentially this analysis will show whether MA or #of missed doses on the day prior is statistically significantly related to seizure on day window+1, while taking into account the fact that some of the data is sampled from the same individuals.

```
lme = fitglme(tbl, 'Seizure_tbl ~ MissedDoses_tbl + MA_tbl + (1|ID_tbl)',  
'Distribution', 'Binomial');  
  
Stats_coeff(iteration,1)=lme.Coefficients{2,2}; %Beta for Missed Doses, each  
row is an iteration  
Stats_coeff(iteration,2)=lme.Coefficients{3,2}; %Beta for MA, each row is an  
iteration  
Stats_p(iteration,1)=lme.Coefficients{2,6}; %p-value for Missed Doses, each  
row is an iteration  
Stats_p(iteration,2)=lme.Coefficients{3,6}; %p-value for MA, each row is an  
iteration  
Stats_CI_MissedDoses(iteration, :) = lme.Coefficients(2, 7:8); % 95% CI for  
MissedDoses  
Stats_CI_MA(iteration, :) = lme.Coefficients(3, 7:8); % 95% CI for MA  
  
% Check for significance and increment counters  
if Stats_p(iteration,1) < 0.05  
    significant_missed_doses = significant_missed_doses + 1;  
end  
if Stats_p(iteration,2) < 0.05  
    significant_ma = significant_ma + 1;  
end  
  
end  
  
% Calculate and display the final proportions  
proportion_missed_doses = significant_missed_doses / 500;  
proportion_ma = significant_ma / 500;  
  
fprintf('Proportion of significant p-values for Missed Doses: %.2f\n',  
proportion_missed_doses);  
fprintf('Proportion of significant p-values for MA: %.2f\n', proportion_ma);
```

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